Case

Patient demographics

Cardiorespiratory Physiotherapy

■ SUBJECTIVE PROBLEM LIST

- Physical Impairments
- Cannot complete household chores without stopping
- Reduced walking distance (likely MRC Grade 3–4)
- Difficulty with personal care (showering, dressing)
- Peripheral muscle deconditioning — ↓ VO■ max
- Social isolation — limited community participation
- Possible anxiety / dyspnoea fear-avoidance cycle
- Unable to speak in full sentences
- Cannot perform ADLs; completely activity-limited
- Unable to attend academic commitments

■■ RED FLAGS

- Haemoptysis
- Blood in sputum in a 74-year-old with 40 pack-year history — must exclude malignancy, PE, or severe infection
- Urgent GP/ED referral; do not commence rehab
- SpO■ <88% at rest
- Severe resting hypoxaemia; exercise will worsen desaturation; may require supplemental O■ assessment (LTOT)
- Refer for ambulatory O■ assessment; modify exercise intensity accordingly
- Ankle oedema + central cyanosis
- Suggests cor pulmonale (right heart failure from chronic hypoxaemia) — medical instability

■ PATHOPHYSIOLOGY (KEY POINTS)

- Fletcher's 40 pack-year history is the dominant aetiological driver.
- Emphysematous destruction causes loss of elastic recoil .
- Without the elastic tethering that normally holds airways open, dynamic airway collapse occurs during expiration — gas is trapped distally.
- Repeated air trapping leads to chronic hyperinflation — the barrel chest.
- The chronically hyperinflated, flattened diaphragm sits at a mechanically disadvantaged position (shorter muscle fibre length, suboptimal force-length relationship on its length-tension curve).
- Breathing therefore recruits accessory muscles — visible at rest in Mrs.

■ DIFFERENTIAL DIAGNOSES

See full case notes

■ OBJECTIVE ASSESSMENTS & HOW TO PERFORM

6-Minute Walk Test (6MWT)

What it measures Submaximal exercise capacity and functional endurance — reflects real-world activity tolerance. Gold standard for pulmonary rehab eligibility and outcome measurement. How to perform

outcome measure for pulmonary rehabilitation and will determine whether

supplemental O■ is required during exercise."

MRC Dyspnoea Scale

What it measures Grades breathlessness severity (1–5) based on the level of activity that provokes it. Validated, simple, and directly used to determine pulmonary rehab eligibility (Grade ≥2). How to perform

"I select the MRC scale because it directly answers

the referral question — is Mrs. Fletcher suitable for pulmonary rehabilitation? It is quick, validated, and provides a grade I can track as an outcome measure at programme completion."

■ TREATMENT PLAN

- Airway Clearance: Active Cycle of Breathing Technique (ACBT)
- Breathing control (relaxed
- tidal breathing, 3–4 breaths, shoulders down) → thoracic expansion
- exercises (3–4 slow deep breaths, optional 3-second inspiratory hold) →
- forced expiration technique / "huff" (open glottis, mid-to-low lung
- volume, not a cough) → repeat cycle 3–4 times → productive cough at end
- REGRESSION: Breathing control + gentle assisted cough only if fatigued; sitting forward-lean position
- PROGRESSION: Add gravity-assisted drainage positions; introduce Flutter/Acapella oscillating PEP device; increase cycles to 4–5
- Breathing Exercise: Pursed-Lip Breathing (PLB) + Diaphragmatic Re-education
- Inhale through nose 2
- counts; exhale slowly through pursed lips ("blow out a candle slowly") 4
- counts. Creates back-pressure (intrinsic PEEP) → slows expiratory flow →